LUMA Cost Model

Formula architecture and implementation blueprint

Focused extraction from Summary, Transportation, Personnel, OH (Overhead), and Result

BOM quantities

Tracker configurations roll into a consolidated item summary.

→

Landed packages

Material, transport, currency and package allocation.

→

Project cost

Personnel, overhead, contingency and finance.

→

Commercial result

Quoted price, fees, penalties, tax and margin.

Purpose: explain the sample workbook faithfully, then translate it into a clean four-package structure suitable for LUMA BOM Manager. No application code or workbook content was changed.

Source	2632-1405_LUMA tracker BOM configurator_rev02 New Price Singsun.xlsx
Sample project	48 trackers | 1,344 PV modules | 720 Wp/module | 967.68 kWp
Extraction scope	5 worksheets | 1,142 formula cells in the selected sheets
Analysis date	31 August 2026

1. Executive architecture

The workbook is a staged cost model, not one monolithic formula.

Configuration sheets

Sheets 15-56 calculate quantities for each tracker arrangement.

→

Summary

Aggregates BOM quantities, weights and item purchase cost.

→

Transport + OH

Transportation creates freight rates; OH feeds Personnel.

→

Result

Builds landed cost, expected cost, quote and commercial margin.

Core dependency chain

Producer	Consumer	What crosses the boundary
Configuration sheets 15-56	Summary	Item quantities by tracker configuration
Summary	Transportation	Steel weight, bearing quantity, slew-drive quantity
OH	Personnel	Annual overhead pool
Summary + Transportation + Personnel	Result	Material, freight, capacity-scaled project cost

Critical finding: quoted price is created independently from expected cost. Margin is calculated afterward as Quoted Price - Expected Cost.

2. What each selected sheet owns

A clean implementation should preserve these responsibilities while moving hardcoded assumptions into managed data.

Sheet	Role in workbook	Main inputs	Main outputs	Recommended LUMA owner
Summary	Consolidated BOM and purchasing basis	Per-configuration quantities, weights, unit prices, currency factor	Quantity, total weight, FOB/Genoa cost by item	BOM engine + Part Master + supplier price list
Transportation	Route and freight allocation	Handling, storage, truck, capacity, item/package weight	Per-kg/per-item freight and package freight totals	Logistics service
Personnel	Annual people pool normalized by project capacity	Monthly personnel cost, annual allocation base, OH annual total	EUR/MW rates and project amounts	Commercial settings + cost engine
OH	Operating-expense ledger	Annual overhead line items	Annual and monthly OH totals	Administration / Overhead settings
Result	Landed package cost and commercial result	Material, freight, people/OH rates, quote lines, toggles	Expected cost, quote, margin, fees and tax	Analysis calculation service

Recommended separation of concerns

Presentation — Analysis UI shows values and assumptions; Administration owns editable commercial data

Calculation — Pure functions calculate item, package, project and commercial totals

Services — Supplier prices, FX, logistics, personnel and overhead revisions

Persistence — Supabase tables with revisions, validity dates, audit fields and RLS

3. Summary: BOM aggregation and item cost

Business formulas are shown with names; Excel references are retained only for traceability.

Output	Business formula	Workbook example	Source reference
Tracker quantity	SUM(tracker quantity across all configuration rows)	48 trackers	Summary T2
PV module quantity	SUM(module quantity across all configuration rows)	1,344 modules	Summary T3
Plant power (kWp)	Module Quantity * Module Rating (Wp) / 1,000	1,344 * 720 / 1,000 = 967.68 kWp	Summary T4
Item quantity	SUM(item quantity across configuration sheets 15-56)	Calculated independently for every TAG	Summary U15:U93
Weight-priced unit cost	Unit Weight * Price per kg	Used for structural steel	Summary Y
Direct-priced unit cost	Supplier Unit Price * FX rate	Slew drive: 208 * 0.87 = 180.96	Summary Y63
Item purchase total	Item Quantity * Item Unit Cost	Slew drive: 48 * 180.96 = 8,686.08	Summary Z
Item total weight	Item Quantity * Unit Weight	Structural package feeds transport	Summary AB

Implementation formula

NormalizedUnitCost[i] = SupplierUnitPrice[i] * FX(PriceCurrency -> QuoteCurrency). For weight-priced items: SupplierUnitPrice[i] means PricePerKg * UnitWeight. ItemMaterialCost[i] = BOMQuantity[i] * NormalizedUnitCost[i].

Control point

Package and subpackage must be stable master-data fields linked to TAG. Do not classify items by searching descriptions. Supplier price revision, currency, incoterm and effective date should remain traceable on every priced item.

4. Transportation: route, capacity and allocation

The workbook mixes per-container, per-kg, per-item and manual fixed allocations.

Cost route	Workbook formula	Sample result	Interpretation
Steel handling/container	400 unload + 400 load + 200 handling + 250 storage	EUR 1,250/container	Storage = 250 * ROUNDUP(15 days / 15-day period)
Steel route/container	Handling/container + warehouse-to-site truck	1,250 + 1,000 = EUR 2,250	Genoa/warehouse to site
Steel freight/kg	Route cost / container capacity	2,250 / 22,000 = EUR 0.102273/kg	Workbook allocates fractionally by actual weight
Steel transport	Steel total weight * freight/kg	36,249.12 * 0.102273 = EUR 3,707.30	Package-weight allocation
Bearing freight/item	Route/container / bearings per container	9,050 / 7,176 = EUR 1.261148/item	192 bearings -> EUR 242.14
Slew local freight/item	Local route/container / units per container	2,250 / 594 = EUR 3.787879/item	48 units -> EUR 181.82

Origin

Supplier price + incoterm defines where responsibility begins.

→

International leg

FOB-to-port, insurance and sea freight when applicable.

→

Warehouse

Unload, storage periods, handling and reload.

→

Project site

Final truck leg allocated by container, kg, item or fixed cost.

Recommended control: store each route leg once with basis_type = per_container | per_kg | per_item | fixed. Then PackageTransport = SUM(applicable leg amount).

5. Personnel and OH: annual pools normalized by MW

The model converts annual company cost pools into a project cost using plant capacity.

Cost pool	Monthly	Annual	Annual allocation base	Rate	Sample project amount
Engineering	EUR 7,824.85	EUR 93,898.20	150	EUR 625.988/MW	0.96768 * 625.988 = EUR 605.76
Test & Commissioning	EUR 14,455.92	EUR 173,471.04	150	EUR 1,156.4736/MW	EUR 1,119.10
Project Management	EUR 9,823.90	EUR 117,886.80	150	EUR 785.912/MW	EUR 760.51
Administration	EUR 38,970.00	EUR 467,640.00	150	EUR 3,117.60/MW	EUR 3,016.84
Office overhead	EUR 24,821.05	EUR 297,852.58	150	EUR 1,985.6839/MW	EUR 1,921.51

Exact model

Stage	Formula
Annual personnel line	MonthlyCost * 12
Annual OH pool	SUM(annual overhead ledger lines) = EUR 297,852.58
Rate per MW	AnnualPool / AnnualAllocationBase
Project amount	ProjectCapacityMW * RatePerMW

The value 150 is the rate denominator but is not explicitly labeled in the workbook. Before implementation, define it as a versioned Annual Allocation Capacity (MW) with owner, unit and valid dates.

6. Result: landed material cost

The Result sheet combines purchasing and freight into commercial cost buckets.

Workbook bucket	Purchase/material	Transport & adjustments	Landed sample	Share of material
Steel Structure	EUR 29,391.12	EUR 16,865.63	EUR 46,256.75	58.47%
Fasteners	EUR 2,130.56	EUR 800.00	EUR 2,930.56	3.70%
Slew Drive	EUR 8,686.08	EUR 849.98	EUR 9,536.06	12.05%
Bearings	EUR 1,927.68	EUR 242.14	EUR 2,169.82	2.74%
SOLTRK	EUR 7,920.00	EUR 800.00	EUR 8,720.00	11.02%
Datalogger	EUR 1,010.82	EUR 400.00	EUR 1,410.82	1.78%
Junction Box	EUR 840.00	EUR 400.00	EUR 1,240.00	1.57%
Safeguard	EUR 3,317.80	EUR 400.00	EUR 3,717.80	4.70%
Accessories	EUR 2,328.00	EUR 800.00	EUR 3,128.00	3.95%
TOTAL	EUR 61,552.06	EUR 17,557.75	EUR 79,109.81	100.00%

General formula

PackageLandedCost[p] = SUM(ItemMaterialCost for package p) + PackageTransport[p] + OtherLandedAdjustments[p]. TotalMaterialLanded = SUM(PackageLandedCost).

Incoterm caveat

The workbook's incoterm toggle suppresses selected Steel and Slew transport adjustments, but it does not suppress the Bearing route or the manually entered local-cost cells. LUMA should define the scope by route leg rather than by one broad Boolean switch.

7. Result: internal cost and expected cost

Capacity-driven project cost is added after landed material.

Line	Formula	Sample
Landed material	SUM(all landed material buckets)	EUR 79,109.81
Engineering	CapacityMW * EngineeringRatePerMW	EUR 605.76
Signature	Manual project service line	EUR 1,000.00
Test & Commissioning	CapacityMW * TestRatePerMW	EUR 1,119.10
Project Management	CapacityMW * PMRatePerMW	EUR 760.51
Administration	CapacityMW * AdminRatePerMW	EUR 3,016.84
Office overhead	CapacityMW * OHRatePerMW	EUR 1,921.51
Finance cost	LandedMaterial * FinanceDays / 365 * AnnualFinanceRate	EUR 0.00
Total project overhead	IF(enabled, SUM(project cost lines), 0)	EUR 8,423.71
Internal cost	LandedMaterial + TotalProjectOverhead	EUR 87,533.52
Contingency	IF(enabled, InternalCost * 5%, 0)	EUR 4,376.68
Expected cost	InternalCost + Contingency	EUR 91,910.19

ExpectedCost = TotalMaterialLanded + ProjectPersonnel + ProjectOH + ProjectServices + FinanceCost + Contingency. Each optional line needs an explicit enabled flag and a visible assumption source.

VAT is not part of the source workbook's Result formula. In LUMA, keep VAT as a separate downstream commercial step: VATAmount = PreVATSellingPrice * VATRate; FinalCustomerPrice = PreVATSellingPrice + VATAmount.

8. Result: quote, margin, fees and tax

The sample quote is a separate commercial build-up, not a cost-plus output.

Quote line	Quantity basis	Rate	Amount
Tracker system	967.68 kWp	EUR 112/kWp	EUR 108,380.16
Safeguards	2	EUR 2,500/item	EUR 5,000.00
Higeco	1	EUR 1,800/item	EUR 1,800.00
Engineering	1	EUR 3,000/project	EUR 3,000.00
Commissioning	1	EUR 4,000/project	EUR 4,000.00
Discount	1	Manual	EUR -9,286.00
Quoted price	SUM(quote lines)		EUR 112,894.16

Commercial result	Formula	Sample
Margin before deductions	QuotedPrice - ExpectedCost	EUR 20,983.97
Margin on expected cost	Margin / ExpectedCost	22.83%
Agent fee	Progressive tiers: 3% <=500k; 2% to 3m; 1% to 10m; 0.5% above	EUR 3,386.82 if enabled
Primary-agent fee	0.5% <=1m; 0.25% above	EUR 564.47 if enabled
Actual margin	Margin - penalties - agent fee - primary-agent fee	EUR 20,983.97 (toggles off)
Tax	IF(enabled, ActualMargin * 27%, 0)	EUR 0.00
After-tax margin	ActualMargin - Tax	EUR 20,983.97 = 18.59% of quote

9. LUMA four-package mapping

This is the requested simplification. Keep subpackages so costs remain explainable.

Package	Subpackages / included items	Recommended classification rule	Cost treatment
Steel Structure	Posts; Substructures	Explicit package + subpackage on every structural TAG	Weight-priced or piece-priced material; freight primarily by kg/container
Major Components	Slew Drive; Bearing; PV Module	Explicit component type; do not infer from description	Direct unit price; freight by item/container/route
Electrical	SOLTRK; Junction Box; Limit Switch; Anemometer; Scada enclosure GW40028; Scada power supply HDR-30-24; Cable gland pg13.5 lapp 53015030; Higeco GWC V4 4DIN; Safeguard-M48V-8-13_rev03; Safeguard-S48V-8-13_rev03	Stable TAG-to-subpackage mapping in Part Master	Direct unit price plus route/fixed allocation
Fasteners	Bolts; Nuts; Washers	Use functional class or approved TAG mapping	Per-piece, per-1000 or package pricing normalized to unit price

Required calculated views

Item

TAG quantity, unit price, FX, weight, material and freight trace.

→

Subpackage

Example: Posts, Bearings, SOLTRK or Bolts.

→

Package

Steel Structure, Major Components, Electrical, Fasteners.

→

Project

Landed material + project costs + commercial result.

The four package totals must always reconcile to the item totals: SUM(ItemLandedCost) = SUM(SubpackageLandedCost) = SUM(PackageLandedCost).

10. Proposed LUMA data architecture

Admin-managed revisions feed a deterministic, read-only Analysis calculation.

Entity	Minimum responsibility	Important fields / controls
Part Master	Engineering identity and classification	TAG, description, unit, weight, package, subpackage, active
Project BOM	Calculated project demand	project_id, TAG, quantity, calculation note, source configuration
Supplier Price Revision	Commercial unit cost	supplier, TAG, category, price, basis, currency, incoterm, valid dates, active revision
FX Revision	Currency normalization	from/to currency, rate, date, source, locked project snapshot
Logistics Route + Legs	Freight and handling	origin/destination, leg, basis type, amount, capacity, storage period, currency
Personnel Rate Revision	Capacity-scaled people cost	cost pool, monthly/annual amount, allocation MW, rate/MW, valid dates
Overhead Revision	Annual OH ledger and allocation	line item, annual amount, allocation MW, valid dates
Quote Settings / Lines	Selling-price construction	strategy, rate basis, quantity source, discounts, VAT, validity, currency
Calculation Snapshot	Auditability and repeatability	input revisions, FX, route assumptions, results, timestamp, user

Security boundary: normal users consume commercial data read-only in Analysis. Admin users maintain revisions in Administration. Supabase RLS, not hidden UI, enforces writes.

11. End-to-end calculation flow

A deterministic pipeline makes every total traceable to its BOM TAG and commercial revision.

1. Freeze inputs

Project BOM, selected supplier revisions, FX date, quote currency.

→

2. Price items

Normalize price basis and currency; calculate item material.

→

3. Land packages

Calculate weight, containers, route legs and freight allocation.

→

4. Add project cost

Capacity-scaled people/OH, services, finance and contingency.

5. Expected cost

Sum four landed packages and all enabled project costs.

→

6. Build quote

Workbook-parity quote lines or explicit future cost-plus strategy.

→

7. Commercial result

Margin, penalties, fees, tax, VAT and final price.

→

8. Reconcile

Item = subpackage = package = project; save calculation snapshot.

Calculation sequence

Order	Invariant / formula
A	ProjectCapacitykWp = ModuleQuantity * ModuleRatingWp / 1,000
B	ItemMaterial = Quantity * NormalizedUnitCost
C	PackageLanded = SUM(ItemMaterial) + PackageTransport + OtherLandedAdjustments
D	ExpectedCost = SUM(PackageLanded) + ProjectCost + Contingency
E	Margin = QuotedPrice - ExpectedCost
F	FinalCustomerPrice = PreVATSellingPrice + VAT

12. Quote strategy decision

Implement workbook parity first; make any future pricing strategy explicit.

Topic	Workbook-parity strategy	Future cost-plus strategy
Quoted price	SUM(independent quote lines + discount)	Derived from Expected Cost and a declared target
Margin	Result of quote versus cost	Input target that drives the quote
Strength	Matches the sample file and existing commercial practice	Consistent target economics
Risk	Quote can drift away from current cost revisions	Confusing markup with margin creates pricing errors
Formula	QuotedPrice = SUM(QuoteLineAmount)	For margin on sales: Quote = ExpectedCost / (1 - TargetMargin)
Alternative	Not applicable	For markup on cost: Quote = ExpectedCost * (1 + MarkupRate)
Recommendation	Phase 1 default	Add later as a separately named strategy; never silently replace parity

Do not label markup as margin. At 20%, ExpectedCost * 1.20 produces a 16.67% margin on sales; ExpectedCost / 0.80 produces a true 20% margin on sales.

For the sample: Expected Cost = EUR 91,910.19 and Quoted Price = EUR 112,894.16. The difference is EUR 20,983.97: 22.83% of expected cost, but 18.59% of quoted sales value.

13. Controls and workbook issues to resolve

These points should be decided before reproducing the model in production code.

Priority	Observation	Recommended decision/control
High	Incoterm toggle affects Steel and Slew adjustments but not Bearing freight or manual local transport.	Apply incoterm responsibility to individual route legs.
High	Quote is independent from expected cost.	Keep parity strategy, show margin warning thresholds, optionally add an explicit cost-plus strategy later.
High	Currency factor 0.87 is embedded directly in the Slew formula.	Use a versioned FX table and save the selected FX revision in the project snapshot.
Medium	Personnel/OH denominator 150 is not explicitly labeled with unit and period.	Store Annual Allocation Capacity (MW/year), owner and validity dates.
Medium	Summary total quantity excludes the final two item rows even though those rows contain quantities.	Replace range totals with full-table aggregation and reconciliation tests.
Medium	Transportation contains an unused helper calculation (48 * 250 = 12,000).	Remove or document it; every formula must feed a named output.
Medium	Freight is allocated using mixed bases and several manual fixed amounts.	Model routes/legs with a declared basis and trace allocations to packages/items.
Low	VAT is absent from the workbook Result sheet.	Keep VAT downstream and visibly separate from cost and pre-VAT quote.

Minimum automated reconciliation tests

Test	Expected invariant
BOM	SUM(configuration quantities by TAG) = consolidated Project BOM quantity by TAG
Packages	SUM(item landed) = SUM(subpackage landed) = SUM(four package landed)
Project	Expected cost = landed packages + project costs + contingency
Quote	Quoted price = SUM(active quote lines); discount preserves sign
Commercial	Margin = quote - expected; actual margin deductions reconcile; VAT is downstream

14. Recommended implementation sequence

Small phases preserve the current engineering/BOM engine and make commercial logic testable.

Phase	Deliverable	Acceptance check
1	Stable package/subpackage mapping in Part Master	Every BOM TAG resolves to exactly one package and subpackage
2	Versioned FX, price-basis normalization and item material cost	Item totals reconcile to supplier revision and quote currency
3	Logistics route legs and allocation engine	Steel, bearing and slew sample freight match workbook within rounding tolerance
4	Personnel and OH revision model	Sample project produces EUR 8,423.71 enabled project overhead
5	Expected-cost result with four-package rollup	Sample expected cost reconciles to EUR 91,910.19
6	Workbook-parity quote lines, fees, tax and VAT	Sample pre-VAT quote reconciles to EUR 112,894.16
7	Calculation snapshot, audit trail and UI explanations	Refresh/reopen reproduces the same inputs and outputs

Sample reconciliation

Checkpoint	Extracted workbook value
Plant capacity	967.68 kWp
Total landed material	EUR 79,109.81
Enabled project overhead	EUR 8,423.71
Internal cost	EUR 87,533.52
Contingency (5%)	EUR 4,376.68
Expected cost	EUR 91,910.19
Quoted price	EUR 112,894.16
Margin before deductions	EUR 20,983.97

Recommended next step: approve the four-package/subpackage mapping and the quote strategy. Then implement the calculation service against these reconciliation values without changing engineering formulas.